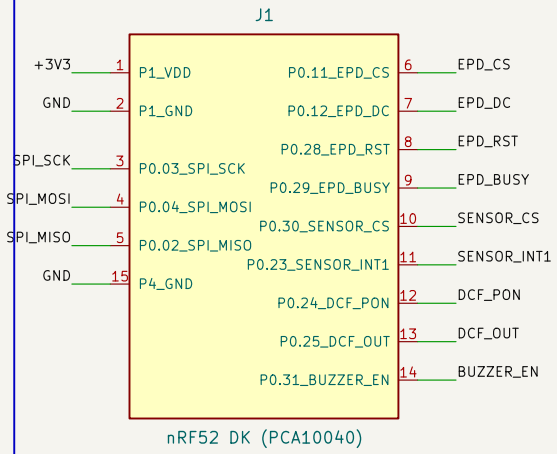
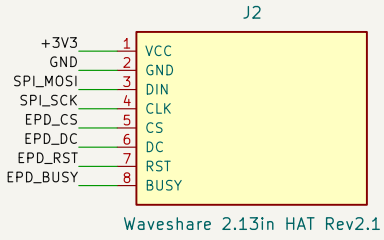
BLOCK 1: HOST CONTROLLER INTERFACE (nRF52 DK / PCA10040)

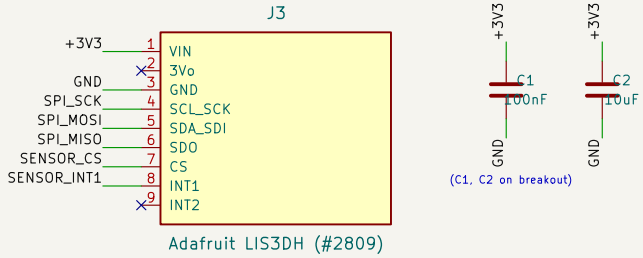


BLOCK 2: DISPLAY INTERFACE (Waveshare 2.13in HAT)

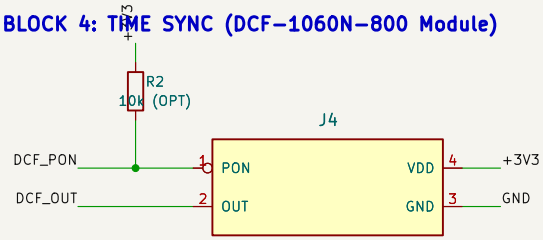


Verify BS1 = 0 (4-wire SPI) on HAT rear switch

BLOCK 3: MOTION SENSOR (Adafruit LIS3DH #2809)

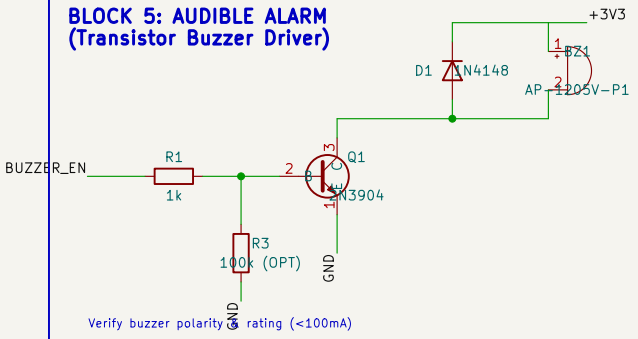


BLOCK 4: TIME SYNC (DCF-1060N-800 Module)



PON = 0V: Enabled (~68 uA) | PON = 3.3V: Disabled (0.0 uA)

BLOCK 5: AUDIBLE ALARM (Transistor Buzzer Driver)



Verify buzzer polarity & rating (<100mA)

nRF52 DK (PCA10040) + E-Paper + LIS3DH + DCF-77 + Buzzer Driver

Bachelor Thesis (APD)

Sheet: /

File: breadboard-prototype-for-promotor.kicad_sch

Title: Ultra-Low-Power Bedside Clock — Breadboard Prototype

Size: A4	Date: 2026-09-13	**Rev: v1.1**
KiCad E.D.A. 10.0.4		Id: 1/1